

DP
rtn
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TUS

Learn DP

gate

Part of DP

Lec_76

Sum of Subsets

Problem Description:-

- Given a set[] of non-negative integers and a value sum, the task is to find there is a subset of the given set whose sum is equal to the given sum.

Ex: Input: set[] = {50, 80, 25, 20, 100, 15, 70}, sum = 250

Output: 1 (Yes) / 0 (No)

Explanation: There is a subset that add up to 250.

[3, 2, 5, 8, 6]
1 2 3 4 5

Sum = 11 (m)
m = 11, n = 5

Sos(11, 5)

take
Sos(5, 4)
Sos(11, 4)

skip
Sos(5, 3)

Sos(0, 2)
Sos(5, 2)

200 + 100 / 0
50 + 80 + 20 + 100
= 250

n = no of elements in set
m = sum that we have to form

Ans = 0 ~ 250

m = 0

$SOS(m, n)$
 if $(m == 0)$
 return 1;

$S[7] = 7$
 (sketch of a DP table)

Algorithm:

if $(n == 0)$ return 0;

if $(S[n] > m)$

return $SOS(m, n-1)$

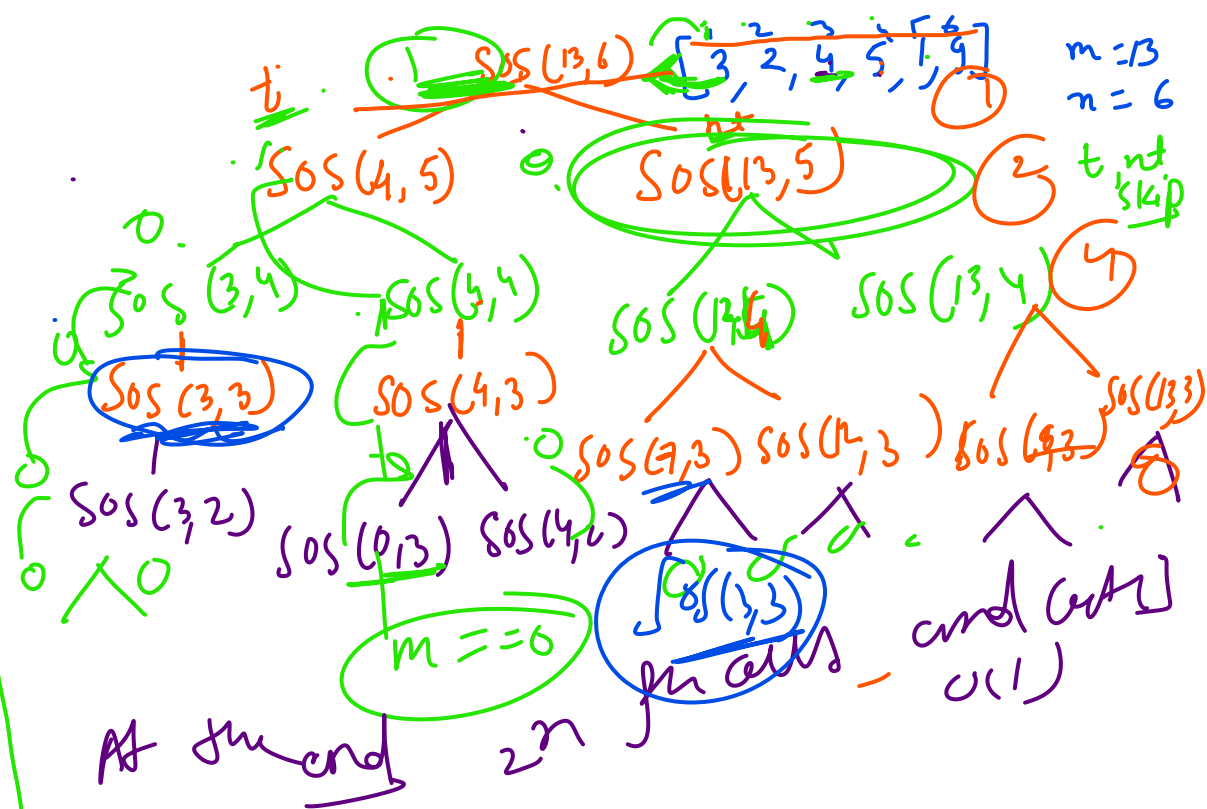
skip

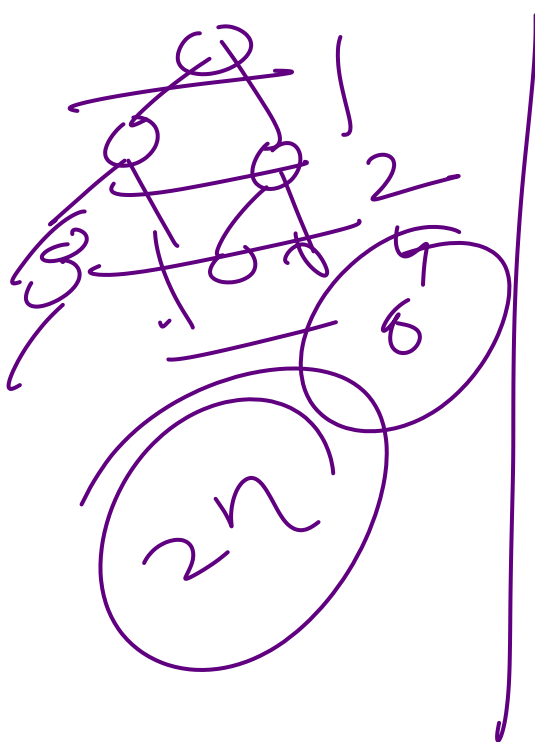
$t = SOS(m - S[n], n-1)$

$nt = SOS(m, n-1)$

return $t || nt$

$BC \approx O(n)$
 $TC = O(2^n) \times O(1) = O(2^n) = O(2^{n+m})$
 (Worst Case)





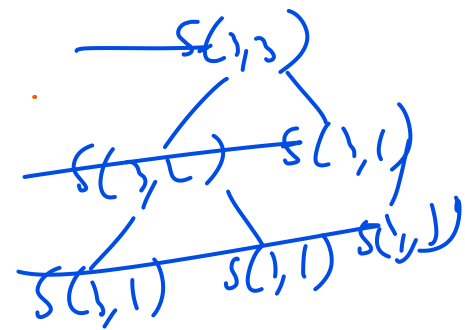
$$\begin{array}{c}
 \text{SOS}(m, n) \\
 \downarrow \\
 \text{SOS}(m, n-1) \\
 \downarrow \\
 \text{SOS}(m, n-2) \\
 \downarrow \\
 \text{SOS}(m, n-3) \\
 \vdots \\
 0
 \end{array}$$

$$\begin{array}{c}
 \underline{O(n)} \quad \text{BC} \\
 \underline{O(2^n)} \quad \text{we} \\
 \text{SC} = \cancel{O(n+m)} \\
 \text{SC} = \underline{O(n)}
 \end{array}$$

Wm DP

SOS(M,n)

$$T(m+1)(n+1) = \{-1\}$$



TC and SC:

- Same as 0/1 knapsack

```

sos(m,n)
{
  if(m == 0) return 1;
  if(n == 0) return 0;
  if(s[n] > m)
  {
    return sos(m, n-1)
  }
  t = sos(m - s[n], n-1)
  nt = sos(m, n-1)
  return t || nt
}

```

$T(m,n)$
 $T(m+1, n+1) = 2^{-1}$
 $\{ \text{if } m = 0 \} = \{ T(m)(n) = 1, \text{ return } T(m)(n) \}$
 $\{ \text{if } n = 0 \} = \{ T(m)(n) = 0, \text{ return } T(m)(n) \}$

```

if (s[n] > m)
{
  if (T(m)(n-1) == -1)
  {
    T(m)(n-1) = sos(m, n-1)
    return T(m)(n-1)
  }
  if (T(m-s[n])(n-1) == -1)
  {
    T(m-s[n])(n-1) = sos(m-s[n], n-1)
  }
  if (T(m)(n-1) == -1)
  {
    T(m)(n-1) = sos(m, n-1)
  }
  return T(m-s[n])(n-1) || T(m)(n-1)
}

```

non DP

$$TC = O(2^n)$$

$$S = n$$

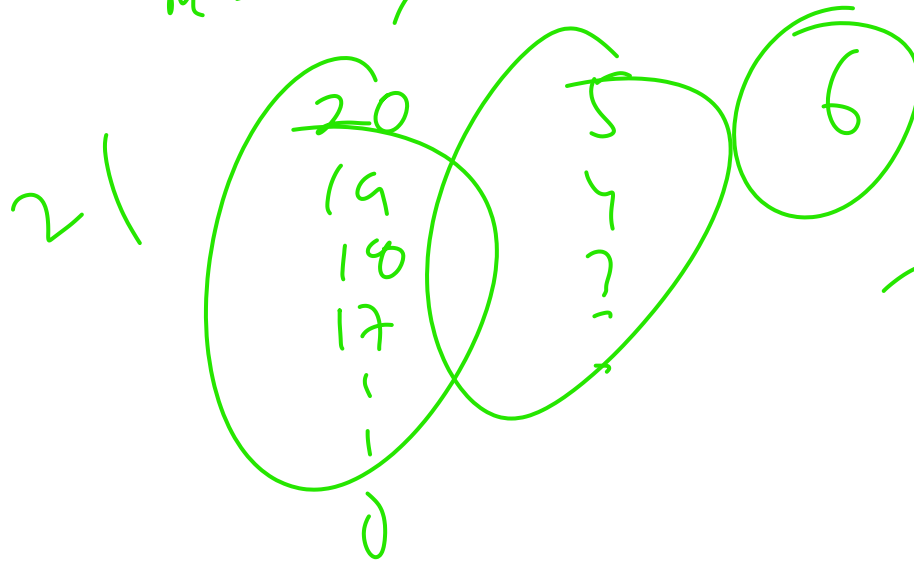
DP

$$O(mn) = \underline{O(4^4)}$$

$$O(mn) = \underline{O(4^4)}$$

Same as d1'

$$m = 20, \quad n = 5$$



21×6

$(m \times n)$

$TC = \underline{O(mn)}$

$O(1)$

